



ChatGPT

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What is ChatGPT?

- ChatGPT is an AI chat box made by OpenAI
- 30th November 2022
- GPT in ChatGPT stands for Generative Pretrained Transformer
- It became so popular, that it had 1 million users in 5 days after going viral on social media
- By comparison Instagram gained 1 million users in 2.5 month

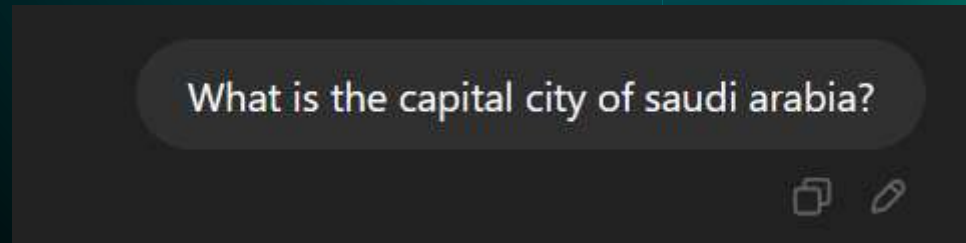


The brains of ChatGPT

- Transformer is the brains of ChatGPT.
- It's trained on large quantity of Data.
- Transformers in ChatGPT only consists of decoders
- decoder helps ChatGPT form sentences and phrases by looking for patterns and how words relate to each other so that it can generate a response.

How does the transformer work?

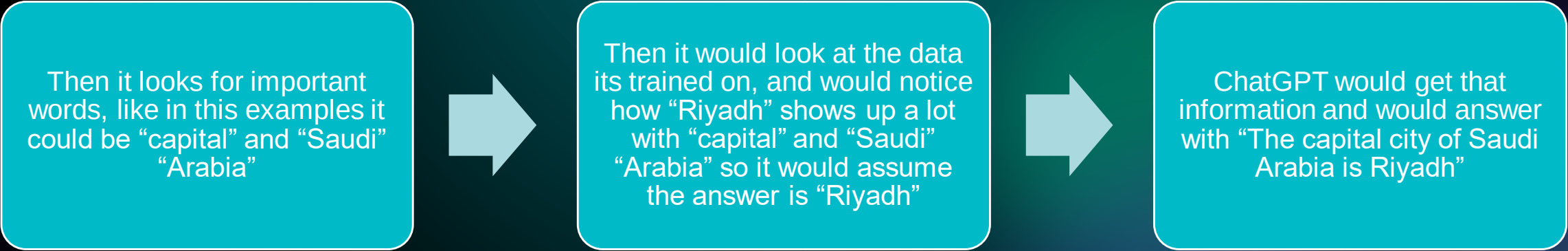
For example, if I ask ChatGPT "What's the capital city of Saudi Arabia?"



The transformer gets the message and splits them into smaller words "What's" "the" "capital" "city" "of" "Saudi" "Arabia" (sometimes what's is split into "what" and "is")

Then it transforms the words into vectors (math) so that the transformer could understand what you wrote

Then it looks for important words, like in this examples it could be “capital” and “Saudi” “Arabia”



```
graph LR; A[Then it looks for important words, like in this examples it could be "capital" and "Saudi" "Arabia"] --> B[Then it would look at the data its trained on, and would notice how "Riyadh" shows up a lot with "capital" and "Saudi" "Arabia" so it would assume the answer is "Riyadh"]; B --> C[ChatGPT would get that information and would answer with "The capital city of Saudi Arabia is Riyadh"];
```

Then it would look at the data its trained on, and would notice how “Riyadh” shows up a lot with “capital” and “Saudi” “Arabia” so it would assume the answer is “Riyadh”

ChatGPT would get that information and would answer with “The capital city of Saudi Arabia is Riyadh”

What is the capital city of saudi arabia?

The capital city of Saudi Arabia is **Riyadh**.

Versions of GPT

- GPT has gone through many versions before it has been ChatGPT we know today
- At first GPT-1 was developed on 2018, followed by GPT-2 and GPT-3
- They weren't popular because you could only access them with an API
- API is like the middleman, you send it a message, it takes your message and sends it to the model (GPT-1...etc.) then it takes whatever answer the model generates and sends it back to you.

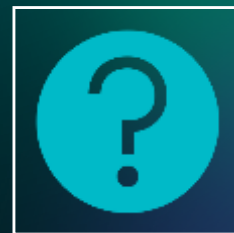
GPT-1



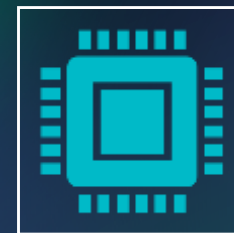
Was the first transformer-based model made by OpenAI



Struggled to understand long conversations



Would often answer in ways that don't make sense or would give irrelevant answers



Access only with API so it was restricted to only researchers and developers

GPT-2



A huge improvement from
GPT-2



Could write essays and do
simple tasks like translating
and summarization



Answers better, and more
human-like



Access was still with an API
so it was limited

GPT-3



Largest Language learning model (LLM) at the time



It could write poetry, articles, essays, code and answer complex questions



Conversations felt more real and human-like



At first it was only accessible through a private API, however OpenAI has made ChatGPT so anyone could access it.



This started gaining peoples attention

GPT-4

Latest model and what ChatGPT is currently on

Most powerful version so far

Could understand images

It,s better at reasoning and problem solving

Could handle even longer conversations

Memory

When you chat with ChatGPT (voice, texting, image generating) it stores information like, what you like, what you dislike and interests.

It does this to so it could respond to you in a more personal manner

There is two main parts of the memory, saved memory and chat history

You can turn on and off your ChatGPT's memory depending on what you like

Saved memory

This is where ChatGPT stores information from you in something called the “notebook”

Information that's been saved are useful and make chatting with ChatGPT more personal

For example; You told ChatGPT you're allergic to nuts, ChatGPT gets this information and stores it in the “notebook”, next time you might ask for a baking recipe, ChatGPT will avoid giving you any recipe with nuts in it.

You can save information manually using “Remember”, e.g. “remember I'm allergic to nuts”

Or ChatGPT will store it automatically when you mention it casually, e.g. “I'm allergic to nuts”

Remember I'm allergic to nuts

Updated saved memory

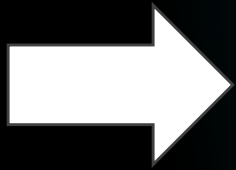
Got it — no nuts for you, ever. Want me to help you find some nut-free snacks or recipes?



Then let's open another Chat log  and ask for a baking recipe

I want to bake, could you give me a baking recipe?

Absolutely! What kind of vibe are you going for—comforting and cozy (like cookies or banana bread), classy and pretty (like a sponge cake or tart), or quick and fun (like brownies or cupcakes)? Want it to be allergy-safe too (no nuts)?



You can delete anything
you want, in two ways



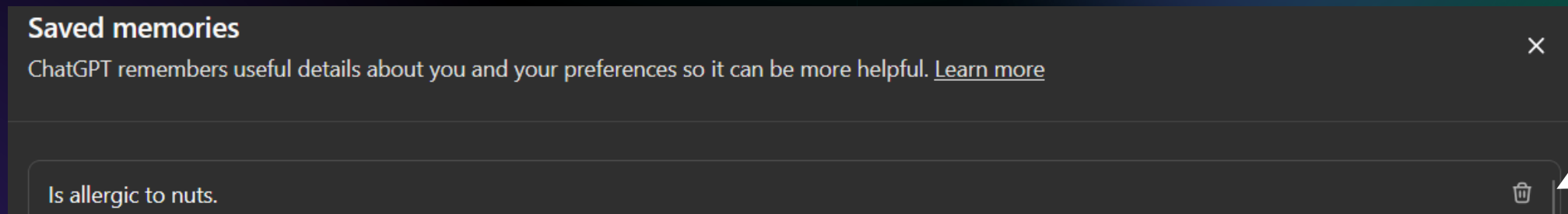
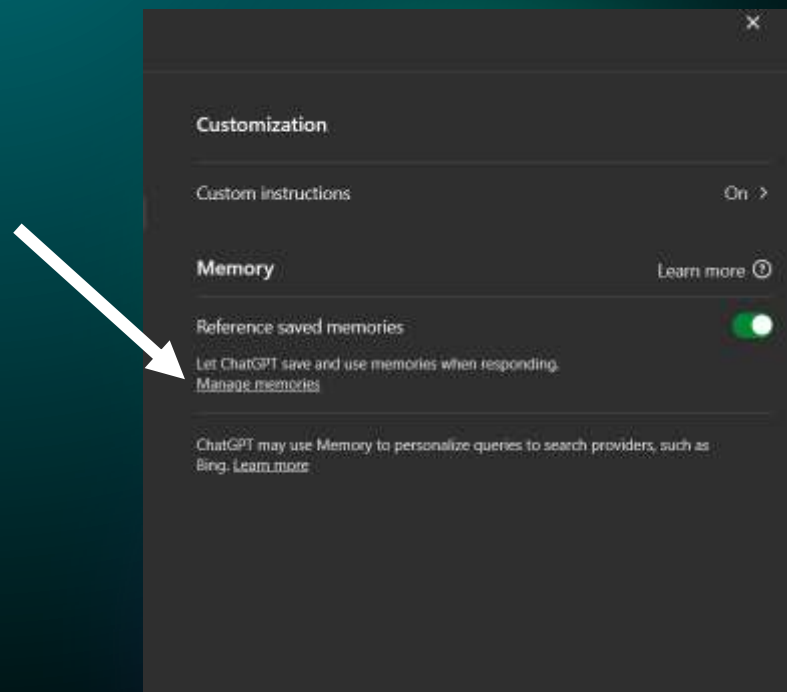
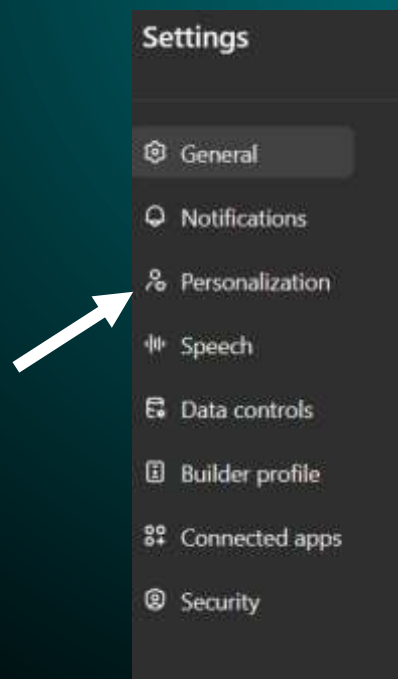
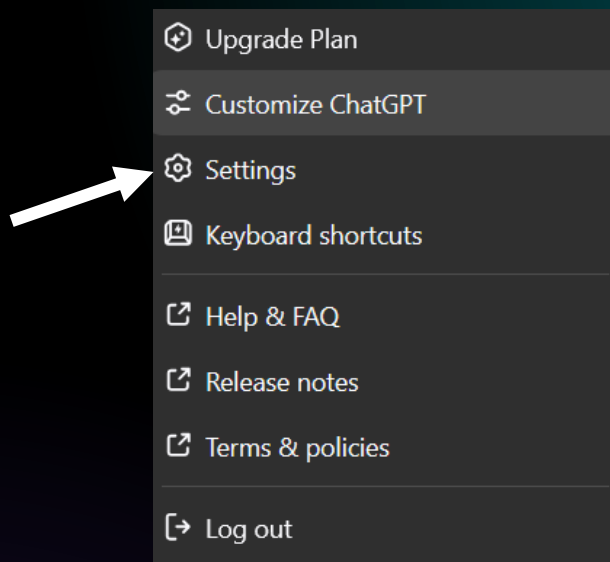
You can ask for a specific
information to be deleted
(doesn't always work)



Or you could do it manually
by clicking on settings >
personalization > manage
memory, then delete



You can turn "Saved
memory" off if you don't
want ChatGPT to
remember anything



Chat History

- This is the memory of the same Chat log, so ChatGPT only remembers if you're talking in one Chat log, if you open another Chat log, ChatGPT won't remember what you said in the previous log

lucy my friend is a nurse, what hairstyle do you reccomand

For your nurse friend Lucy, a great hairstyle would be cute, low-maintenance, and secure enough for long shifts. Here are some perfect nurse-friendly hairstyles:

✂ Practical & Cute Nurse Hairstyle Ideas:

1. Low Bun (with or without a braid):

- Keeps hair off the neck
- Professional and comfortable
- Add a side braid or twist for flair

2. Braided Ponytail:

- Stays put all shift
- Prevents tangling and frizz
- Looks stylish even after 12 hours

• Looks stylish even after 12 hours

• Prevents tangling and frizz

what was lucy's job?

Can you clarify which Lucy you mean? There are a few famous Lucys with notable jobs, such as:

- **Lucy Pevensie** from *The Chronicles of Narnia*: She was a Queen of Narnia.
- **Lucille Ball** (often called "Lucy"): She was a famous actress and comedian, best known for *I Love Lucy*.
- **Lucy Ricardo** (a character played by Lucille Ball): Her "job" was often trying to break into showbiz, though she was technically a housewife.
- **Lucy** from the movie *Lucy* (2014, played by Scarlett Johansson): She was a student who accidentally gained superhuman abilities.



Open another Chatlog

Privacy concerns

ChatGPT collects data all the time, memories, chat logs, conversations.

Because OpenAI want to train ChatGPT to be more human-like, so get data from their users

OpenAI has tried to use synthetic data (data made by AI) but this usually gets weird and strange after a while

Italy has fined OpenAI 15 million euros (64,048,500 ryals) for not being clear about the users' data being collected



The more data OpenAI collects the more data is stored



Hacked/breached → identity theft/scams



March 2023 there was a data leaked, and everyone's information was been shown publicly



Google researchers have found a way to make ChatGPT give them the training data (which includes users' data)



When you sign in to ChatGPT using your email address, it collects information about your IP address, information about your device and your location whenever you use its service



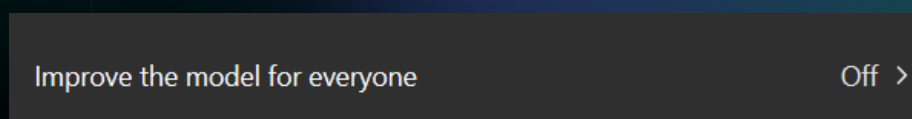
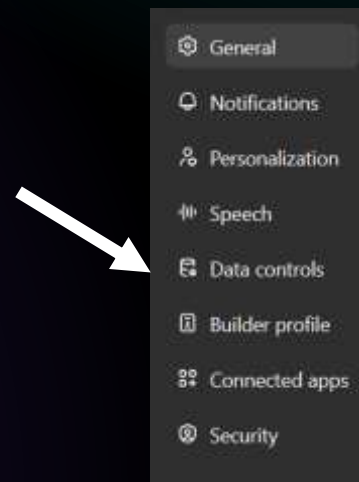
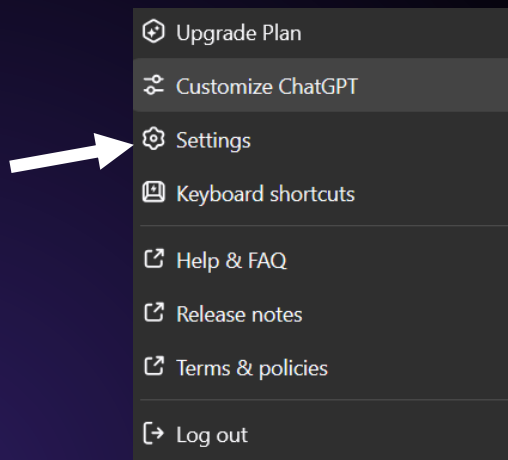
In 2023, quarter of a million of users' login credentials were sold on the black market

Staying safe on ChatGPT

Here are ways to stay safe on ChatGPT;

- Never share credit information, address, close restaurants, phone numbers, passwords, etc.
- It's safer to use ChatGPT without your account
- In settings close the "Improve the model for everyone"
- Wipe any memory that is too personal from "Saved memory"
- You could also use "temporary chat" which is a chat that OpenAI doesn't collect your data from, and it wipes anytime you open a new chat log

 Temporary



Misinformation

- Many people use ChatGPT like google now, because it gives answers much faster and much easier, you don't need to spend hours researching, but should we rely on ChatGPT for true information?



Google VS ChatGPT

- Google
- Collects databases and give you links
- Information will be as it is written
- Has reliable links
- User could distinguish between an opinion and a fact

- ChatGPT
- Collects databases (from internet, training data, etc) and give you answers based on a series of guesses
- You don't know where ChatGPT pulled this information from
- Can't distinguish between a fact and an opinion
- If it doesn't know the answer it will confidently give you false information, and when you correct it, it will argue

Did you know?

- ChatGPT is 62% more likely to give you misinformation
- ChatGPT has been trained on data that's before 2021, so information could be outdated.

➔ Information by ChatGPT-like google, should be checked first before believing or using

Mental health

ChatGPT could be a great tool for users' mental health, using "mental health GPT"

- Improve accessibility
- Staying anonymous
- Long time support

It could also have a negative impact

- ChatGPT is really human-like, might cause social isolation → depression
- ChatGPT, helps it doesn't diagnose
- ChatGPT has biases and inequity, might cause anger

Impact on education

ChatGPT has been helping students with exams, essays and homework and educators are worried

Bad impacts of Chat;

- Weaken critical thinking
- Reduces students' ability to learn independently
- Could harm students' original thoughts and their creativity
- A survey conducted by best colleges in October 2023, has found that 43% of the students that participated have admitted to using AI tools-like ChatGPT. Nearly 50% of these students also confessed to use AI (like ChatGPT) for exams and assignments

Its not all bad, if used responsibly ChatGPT could a be great tool, some good potential impact of ChatGPT on education;

- Makes education more accessible (translation, reading out loud for the blind etc.)
- Helps prepare mock exams (generates practice question for the student to practice)
- Personalized learning (ChatGPT could understands the users learning style and help them study accordingly).
- Helps with homework
- Clarify doubts a student might have on a certain topic.

Impact on the environment

ChatGPT isn't good for the environment

- According to estimates, ChatGPT emits 8.4 tons (8500kg) of carbon dioxide per year
- Research at university of California, Riverside done by Shaolei Ren and his team which included Pengfei Li and Jianyi Yang, and Mohammad A. Islam, stated that AI models like ChatGPT have a really high water footprint- which is the amount of water used to produce goods- ,the study revealed that Microsoft has used 700000 Liters of water when training ChatGPT-3's data centre

A simple 50 question conversation with ChatGPT needs 500ml of water

Why do you think ChatGPT use water, when its traing or generating texts for users?

Because when training, large amounts of energy is used, which produces heat and would need large amounts of water to cool down. Lots of water is used when it process users input and generates an output

Does ChatGPT have a conscious?

- First of all, how would you describe a conscious?
- Philosophers say for a thing to be conscious it needs to have individual experiences, goals and beliefs
- So, does ChatGPT have a conscious?

- A study by Dr. Clara Colombatto who's a psychology professor and her assistant Dr. Steve Fleming who's a professor of cognitive neuroscience, have conducted a survey of 300 U.S participants. They've found that two thirds of the participants believe ChatGPT has a conscious. Colombatto has said it was really surprising that people have thought that way, as ChatGPT is mainly only texts and a voice with no face. She found that the more people used ChatGPT the more likely that think its conscious because people have come to trust it which, by the way, is exactly what innovators want, because if people trust their product they'll use it often. Do you believe ChatGPT is conscious?

The co-founder of OpenAI, Ilya Sutskever has said the algorithms behind ChatGPT might be "slightly conscious" as he put it.

Stephen Wolfram who's a computer science has argued that no, ChatGPT isn't conscious in his website, he says it uses math to figure out what word to use in different contexts based on the data it's been trained on

The founding director of Florida Atlantic University's Centre for the Future Mind, Susan Schneider said that current chat box, like ChatGPT use human data and writings, so to test if a program is conscious, they'd have to not give it access to any of that and see if it can describe its experiences.

So, in short, while ChatGPT doesn't have a conscious, it does act like and interact like that, that said let's move on a different topic, features in ChatGPT.

Features

ChatGPT isn't just chatbox

Canvas;

- Websites, Windows, and MacOS
- You cant access Canvas directly
- ChatGPT opens it when writing a code or to write
- Opened automatically when ChatGPT senses it needs it
- Opens manually with "open a canvas"

open a canvas and write an essay on the internet

Internet Essay

The Internet: A Double-Edged Sword

The internet has become one of the most transformative inventions in human history. It has revolutionized the way people communicate, access information, conduct business, and even form relationships. However, like most powerful tools, the internet is a double-edged sword that brings both incredible benefits and significant drawbacks.

On the positive side, the internet has democratized information. With just a few clicks, anyone with an internet connection can access vast amounts of knowledge on virtually any topic. This has empowered individuals to learn new skills, stay informed, and pursue educational opportunities that were once limited to formal institution. Typing ... tforms like YouTube, Khan Academy, and

write a python code to make a calculator

Heck yesss let's make you a cute lil' calculator in Python! Here's a basic one that can do addition, subtraction, multiplication, and division:

```
python
def calculator():
    print("Welcome to Baby's First Calculator 🧮💖")

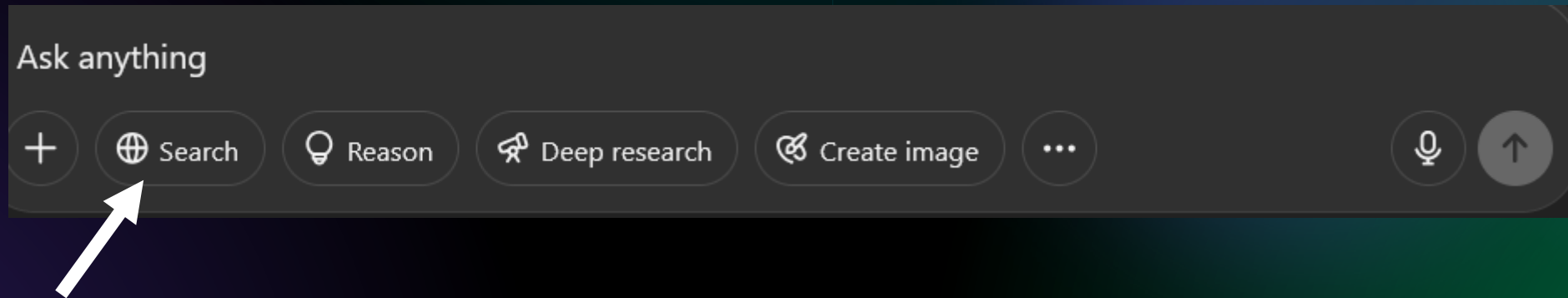
    while True:
        print("\nChoose operation:")
        print("1. Add (+)")
        print("2. Subtract (-)")
        print("3. Multiply (*)")
        print("4. Divide (/)")
        print("5. Quit")

        choice = input("Enter your choice (1/2/3/4/5): ")

        if choice == '5':
            print("Okay byeeee 🍀💖")
            break
```

Web browsing;

- Because its trained only on data before 2021, you could make ChatGPT search on the internet to give you the latest news or fact
- Manually "search for.."
- Automatically, you could ask it a question and it could sense it needs to search about it so it searches



search about the moon

Searching the web

what are some 2030 saudi visions

Searching the web

Deep searching;

- extends normal searching
- In depth search with multiple sites
- Limit for free ChatGPT users, 5 times every 30 days

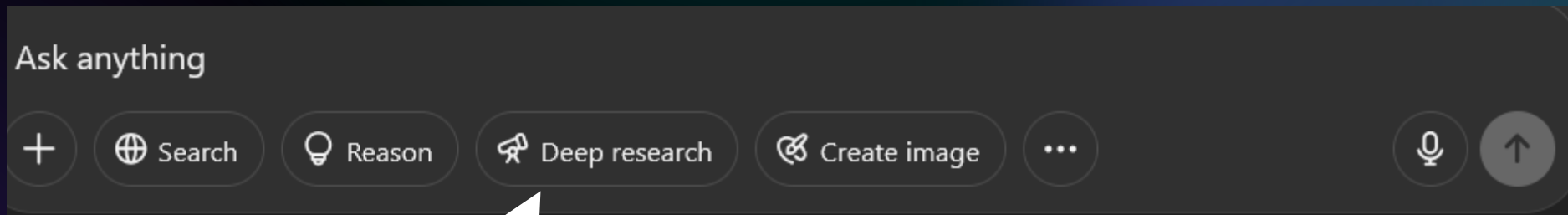
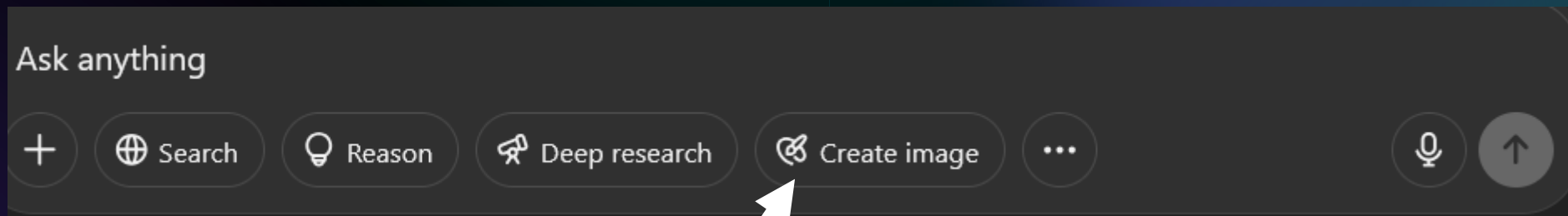


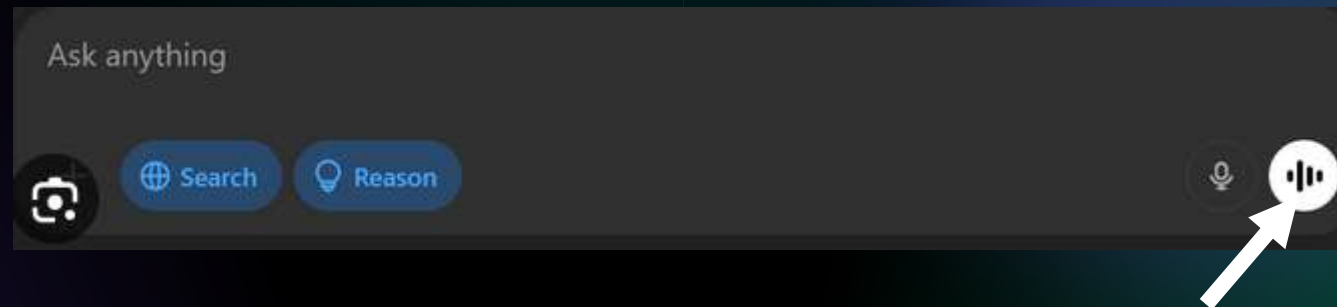
Image generating and processing;

- Uses GPT vision mode
- Can upload an image and ask ChatGPT questions about it
- ChatGPT can generate an image based on what you describe, it doesn't need to be detailed
- "generate an image.."]
- Click on "create an image"



Voice mode;

- Talk with ChatGPT through your voice, it'll respond with a voice
- They sound very human like, with a pitch and everything
- Only with app
- Helps with English learning



THANK YOU FOR LISTENING

Reference

[1] <https://www.emerald.com/insight/content/doi/10.1108/lhtn-01-2023-0009/full/html>

[2]<https://www.forbes.com/sites/bernardmarr/2023/05/19/a-short-history-of-chatgpt-how-we-got-to-where-we-are-today/>

[3] <https://uca.edu/cetal/chat-gpt/>

[4][6]<https://www.wired.com/story/how-chatgpt-works-large-language-model/>

[5]<https://biodatamining.biomedcentral.com/articles/10.1186/s13040-023-00339-9>

[7]<https://towardsdatascience.com/meet-gpt-the-decoder-only-transformer-12f4a7918b36/>

[8] <https://www.techbusinessnews.com.au/blog/chatgpt-may-lead-to-the-downfall-of-eduction-and-critical-thinking/>

[9] https://www.bestcolleges.com/research/college-students-ai-tools-survey/?utm_source=chatgpt.com

[10] <https://www.hurix.com/blogs/chat-gpt-pros-and-cons-of-using-chatgpt-in-higher-education/>

[11] <https://help.openai.com/en/articles/8590148-memory-faq>

[12] <https://pirg.org/edfund/resources/chatgpt-privacy/>

[13] <https://apnews.com/article/italy-privacy-authority-openai-chatgpt-fine-6760575ae7a29a1dd22cc666f49e605f>

[14] <https://foundation.mozilla.org/en/privacynotincluded/articles/how-to-protect-your-privacy-from-chatgpt-and-other-ai-chatbots/>

[15] <https://www.mcafee.com/blogs/internet-security/chatgpts-impact-on-privacy-and-how-to-protect-yourself/>

[16] <https://www.frontiersin.org/journals/human-dynamics/articles/10.3389/fhumd.2023.1289255/full>

[17] <https://pmc.ncbi.nlm.nih.gov/articles/PMC10867692/>

[18] <https://www.business-reporter.co.uk/ai--automation/chatgpt-the-ultimate-spreader-of-fake-news>

[19] <https://earth.org/environmental-impact-chatgpt/>

[20] <https://themarkup.org/hello-world/2023/04/15/the-secret-water-footprint-of-ai-technology>

[21] https://subjectguides.uwaterloo.ca/chatgpt_generative_ai/falseoutdatedinfo

[22] https://uwimprint.ca/tool-or-conscious-entity-new-study-by-uw-professor-explores-surprising-views-on-ai-like-chatgpt/?utm_source=chatgpt.com

[23] https://www.nbcnews.com/tech/tech-news/chatgpt-ai-consciousness-rcna71777?utm_source=chatgpt.com

[24] <https://help.openai.com/en/articles/9930697-what-is-the-canvas-feature-in-chatgpt-and-how-do-i-use-it>

[25] <https://help.openai.com/en/articles/9260256-chatgpt-capabilities-overview>

[26] <https://help.openai.com/en/articles/10500283-deep-research-faq>

[27] <https://help.openai.com/en/articles/8932459-creating-images-in-chatgpt>